

House Price Prediction Using Linear Regression

Sumerix Global AI & Machine Learning Internship – Task 1

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1. Objective

The objective of this project is to build and evaluate a Linear Regression model using the California Housing Dataset. This project demonstrates the complete machine learning workflow including data loading, exploratory data analysis, model training, prediction, evaluation, and visualization.

2. Dataset Description

The California Housing Dataset is a popular dataset available in Scikit-learn. It contains information about housing districts in California.

Features Used

Median Income (MedInc)

House Age (HouseAge)

Average Rooms (AveRooms)

Average Bedrooms (AveBedrms)

Population

Average Occupancy (AveOccup)

Latitude

Longitude

Target Variable

Median House Value (MedHouseVal)

The dataset was loaded directly using Scikit-learn and analyzed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify relationships between variables.

Observations

No missing values were found in the dataset.

Median Income showed the strongest positive correlation with house prices.

House prices were not perfectly normally distributed.

Correlation analysis helped identify the most influential features.

Correlation Heatmap:

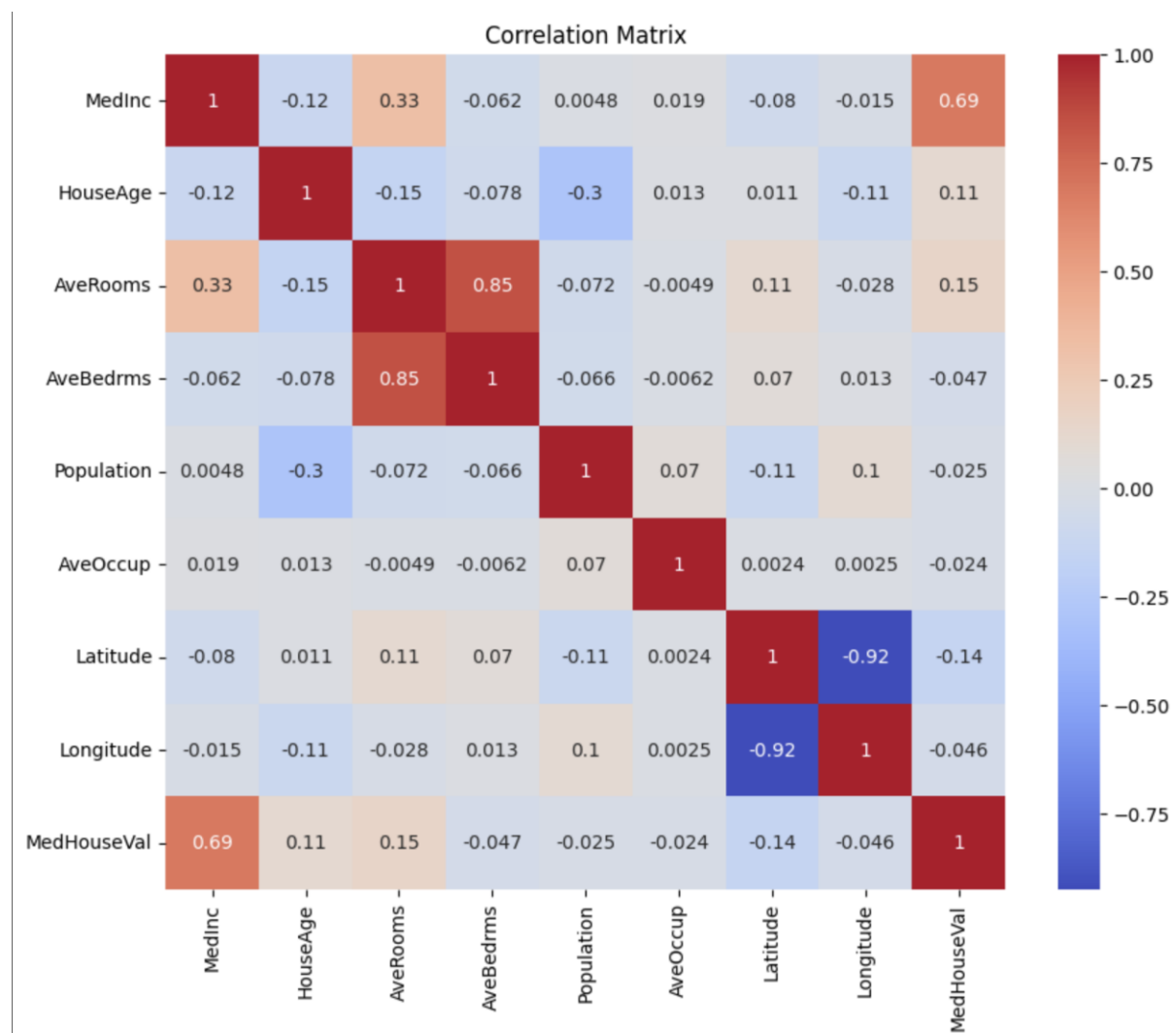


Fig 1 Correlation Heatmap

House Price Distribution

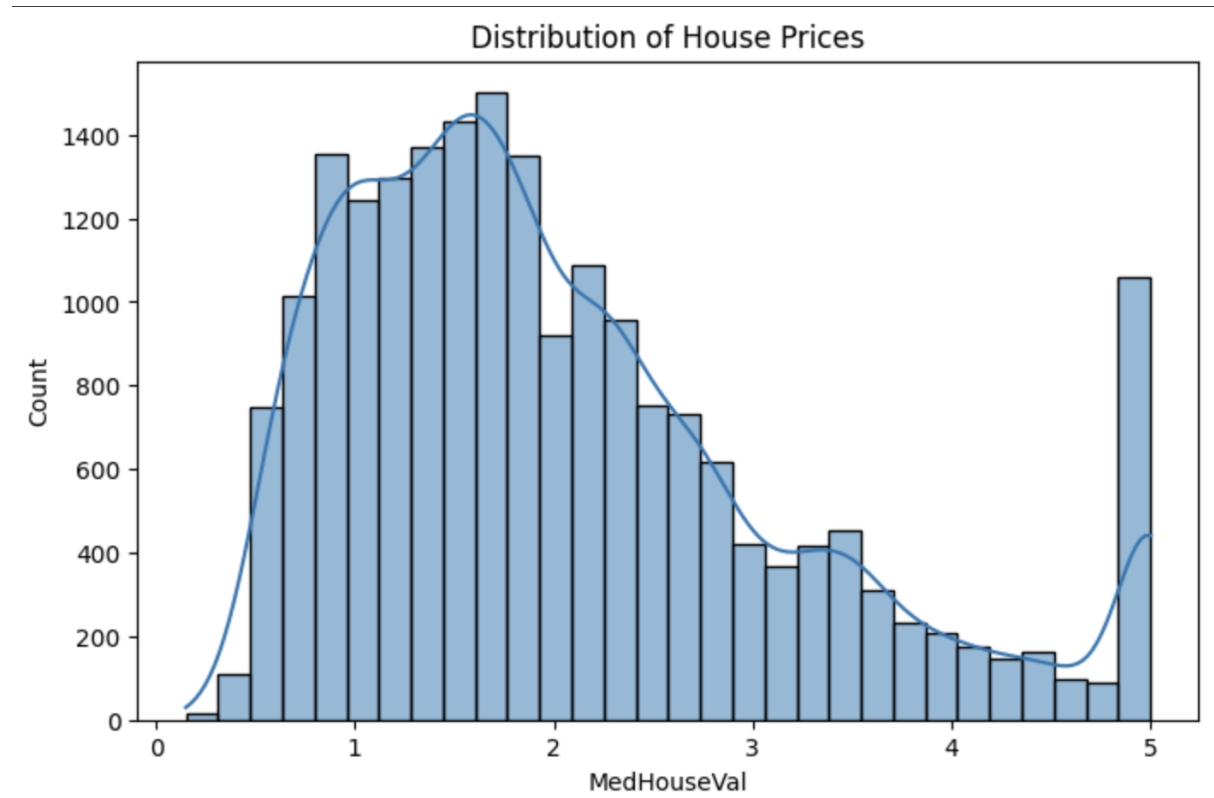


Fig 2 House Price Distribution

4. Model Development

A Linear Regression model was developed using Scikit-learn.

Steps Performed

Data Loading

Data Exploration

Train-Test Split (80:20)

Model Training

Prediction

Evaluation

The model was trained on 80% of the data and tested on the remaining 20%.

5. Model Evaluation

The model performance was evaluated using the following metrics:

Results

MAE = 0.5332001304956555

RMSE = 0.7455813830127763

R^2 Score = 0.5757877060324508

Actual vs Predicted Values

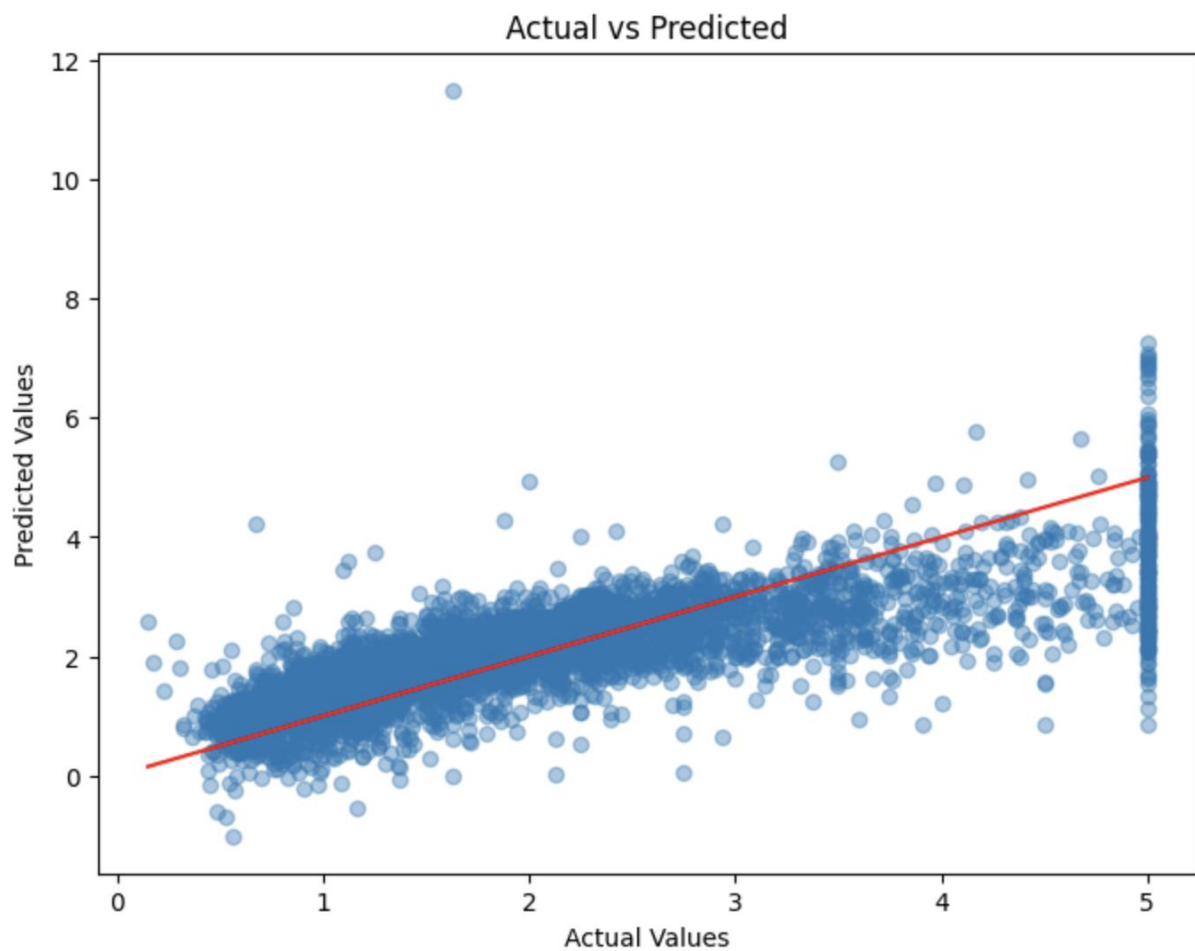


Fig 3 Actual Values vs Predicted Values

The model demonstrated moderate prediction accuracy and successfully captured the relationship between housing features and house prices.

6. Conclusion

This project successfully implemented a Linear Regression model for house price prediction using the California Housing Dataset. Exploratory Data Analysis helped identify important relationships among features, while evaluation metrics provided insights into model performance.

The project demonstrated the complete machine learning workflow from data preprocessing to model evaluation.

7. Future Improvements

- Feature Engineering
- Hyperparameter Tuning
- Random Forest Regression
- XGBoost Regression
- Cross Validation